Topics to discuss

→ Trapezoidal Rule

→ Numerical Problem

→ Error in Integration.

→ Homework Problem

Trapezoidal Rule :

$$\int_{x_0}^{x_n} y\,dx = \frac{h}{2}\left[y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n\right]$$

This is known as the trapezoidal Rule.

The geometrical significant of this rule is that the curve $y=f(x)$ is replaced by n straight lines joining the points (x_0, y_0) and (x_1, y_1) ; (x_1, y_1) and (x_2, y_2) .... ; (x_{n-1}, y_{n-1}) and (x_n, y_n). The area bounded by the curve $y=f(x)$, the ordinates $x=x_0$ and $x=x_n$, and x-axis is then approximately equivalent to the sum of the areas of the n trapezium obtain.

Numerical Problem :

Q: Evaluate approximately, by Trapezoidal rule, the integral $\int_0^1 (4x - 3x^2)\,dx$ by taking $n = 10$.

Solution: length of interval will be,

$$h = \frac{b-a}{n} = \frac{1-0}{10} = 0.1$$

Let, $f(x) = 4x - 3x^2$

Now, Lets make the table for $f(x) = 4x - 3x^2$

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
$f(x)$	0	0.37	0.68	0.93	1.12	1.25	1.32	1.33	1.28	1.17	1

Now by Trapezoidal Method,

$$\int_0^1 f(x)\,dx = \frac{h}{2}\left[y_0 + 2(y_1 + y_2 + y_3 + y_4 + \cdots y_9) + y_{10}\right]$$

$$= \frac{0.1}{2}\left[0 + 2(0.37 + 0.68 + 0.93 + 1.12 + 1.25 + 1.32 + 1.33 + 1.28 + 1.17) + 1\right]$$

$$= 0.05 \times 19.9$$

$$= 0.995 \text{ Ans.}$$

$\therefore$ Approximate value of $f(x) = 0.995$

True value by integration,

$$\int_0^1 (4x - 3x^2)\,dx = \left[4\frac{x^2}{2} - \frac{3x^3}{3}\right]_0^1 = \left[2x^2 - x^3\right]_0^1$$

$$= [2 - 1 - 0] = 1$$

$\therefore$ Absolute error $= 1 - 0.995 = 0.005$

$\therefore$ Relative error $= \left|\frac{1 - 0.995}{1}\right| = 0.005$

Homework Problem

Q: Evaluate $\int x e^x dx$ where the interval $(-1, 0)$ by using Trapezoidal rule taking $n = 6$, correct upto four decimal places.

Solution: Let, $f(x) = xe^x$

and, $h = \frac{0-(-1)}{6} = \frac{1}{6} = 0.1667$

Now, Lets make the integral table,

we have, $y = f(x) = xe^{x}$

x	-1	$-\frac{5}{6}$	$-\frac{4}{6}$	$-\frac{3}{6}$	$-\frac{2}{6}$	$-\frac{1}{6}$	0
$f(x)$	-0.36788	-0.36216	-0.342278	-0.30326	-0.23884	-0.14108	0

so by trapezoidal rule,

$$\int_{-1}^{0} f(x)\,dx = \frac{h}{2}\left[y_0 + 2(y_1 + y_2 + \dots y_5) + y_6\right]$$

$$= \frac{1}{6\times 2}\left[-0.36788 - 2(1.387618) + 0\right]$$

$$= -0.2619 \quad \text{(correct upto 4 decimal places)}$$

Connect with me

Start Practicing

i._am._arfin

Arfin Parween

Arfin Parween (Subscribe here too)

Arfin Parween

Arfin Parween